



The Power of Quantum Computing: Swift Sudoku Solver

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Sudoku Solver

We present an example of using a quantum search algorithm to solve a Sudoku puzzle.

Here are our Sudoku Solver constraints:

The Board is a 2x2 matrix

The Board consists of binary digits

The Board must not have conflicting digits in a row or column, like traditional sudoku

V_0	V_1
V_2	V_3

We begin the **search process** for possible solutions by **superposing** the possible qubit values (0 OR 1) for each V onto qubits. Once that is complete then we can run an **Oracle** on the cubits with the set of rules stated. That way we can mark the state/s that we want to find. Lastly we implement an **Amplitude Amplification Circuit** and measure the cubits to get a result

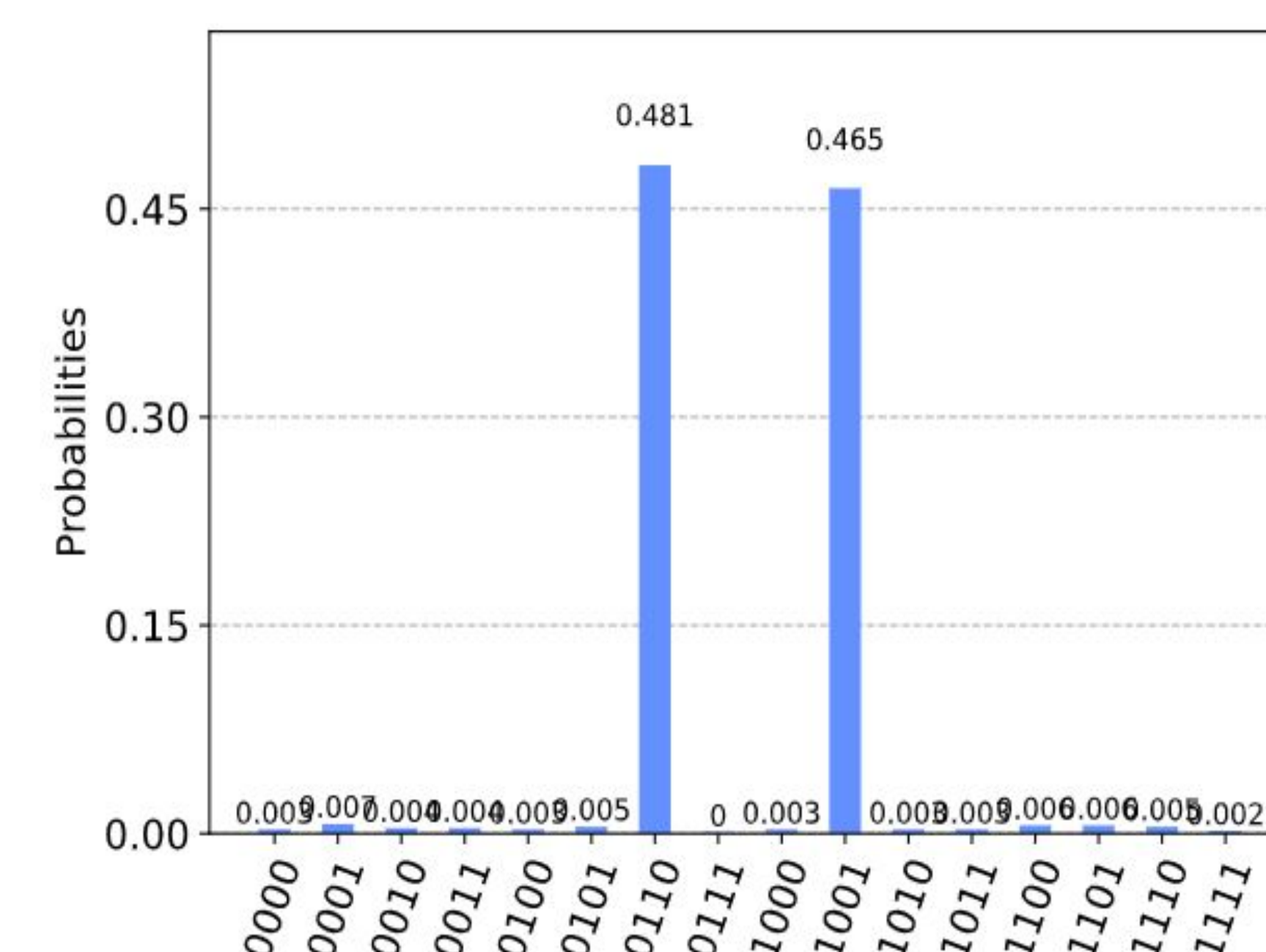


Figure 2

The Sudoku Solving algorithm utilizes a *phase kickback*, uncomputation to turn a quantum circuit into an oracle, then Grover's algorithm is used to solve the oracle. This is what Figure 1 displays

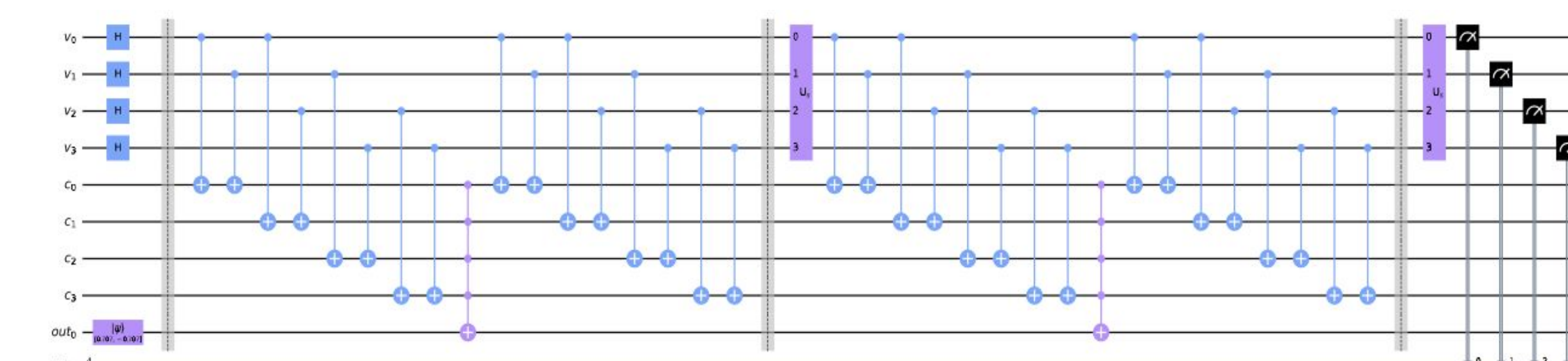


Figure 3

The above is an output of what the quantum circuit looks like after executing the full sudoku solving algorithm. When $|out_0\rangle$ is 1, then that means that there is a solution for the sudoku problem, and we see that this is the case. The qubits are returned to its original state $|0000\rangle$ as indicated by the right side where the circuit is computed again to perform "uncomputation" to make this happen. Thus, we have verified there's a solution to this sudoku.

Understanding The Basics

Quantum computers are vastly different from classical computers. Classical computers use bits that represent either 0 XOR 1 to perform calculations while quantum computers use qubits that represent 0 OR 1 to perform calculations. It is possible for qubits to be 0 and 1 due to quantum superposition properties. This property along with qubit properties allow quantum computers to compute more complex expressions more efficiently than a classical computer can. Although Quantum Computers are currently limited by technology, several algorithms and process have developed for them already to illustrate their power.

What is Grover's Algorithm?

Grover's algorithm is a quantum unstructured search algorithm created by Lov Grover in 1996. Grover's algorithm runs in $O(\sqrt{N})$ time compared to $O(N)$ in classical computing.

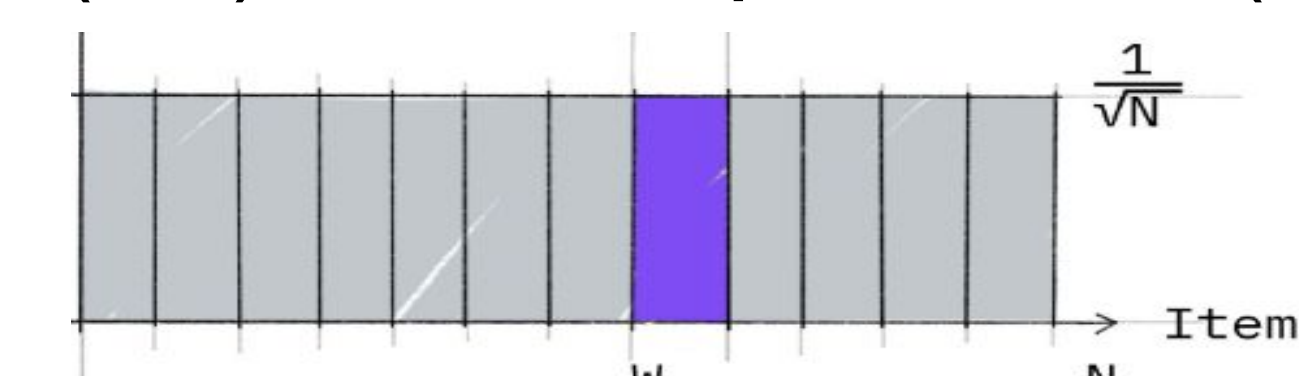


Figure 4

Say we are looking for ω (the purple), which is the item to find, then compared to a classical algorithm $N/2$ steps, we can do it in $\sqrt{N}$ steps. To this we can build a quantum circuit that can output this search using the algorithm.

Grover's Algorithm Process

- 1) Apply a Hadamard gate on the qubits
- 2) Implement an Oracle -> marks the state you want to find
- 3) Implement an amplification circuit
- 4) Measure the qubits -> get result.

References:

1. Qiskit.org [Grover's Algorithm](#)

Acknowledgements

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Sudoku Oracle Formula

$$U_{\omega}|x\rangle|0\rangle|out_0\rangle = \begin{cases} |x\rangle|0\rangle|out_0\rangle & \text{for } x \neq \omega \\ |x\rangle|0\rangle \otimes X|out_0\rangle & \text{for } x = \omega \end{cases}$$

Figure 5

This formula is used in the backend as a function to implement an oracle for this sudoku solving algorithm. U_{ω} is defined as the quantum circuit, x is one of the sudoku variables ($v_0 - v_3$), $|0\rangle$ is the abbreviated qubit clause, $|out_0\rangle$ is the qubit used to check the output of the checking circuit, and X is any arbitrary clause qubit from the quantum circuit. This formula creates the oracle.

Amplitude Amplification

In short, estimation is based upon element with highest probability. The target element's probability is increased with amplitude amplification so it can be found more often

1. Amplitude Amplification begins with uniform superposition in state S. Refer to Figure 1

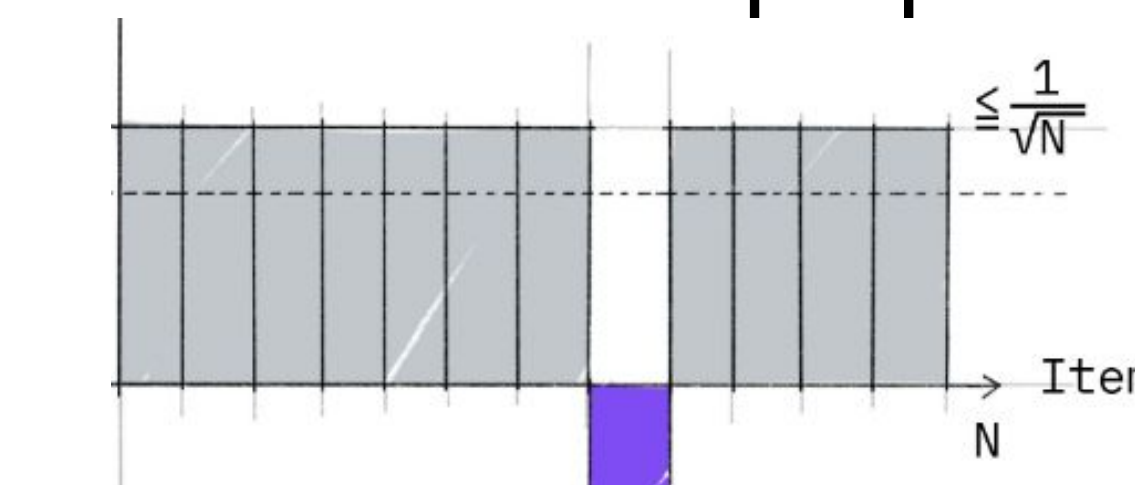


Figure 6

2. An oracle reflect U_f is applied to the state S

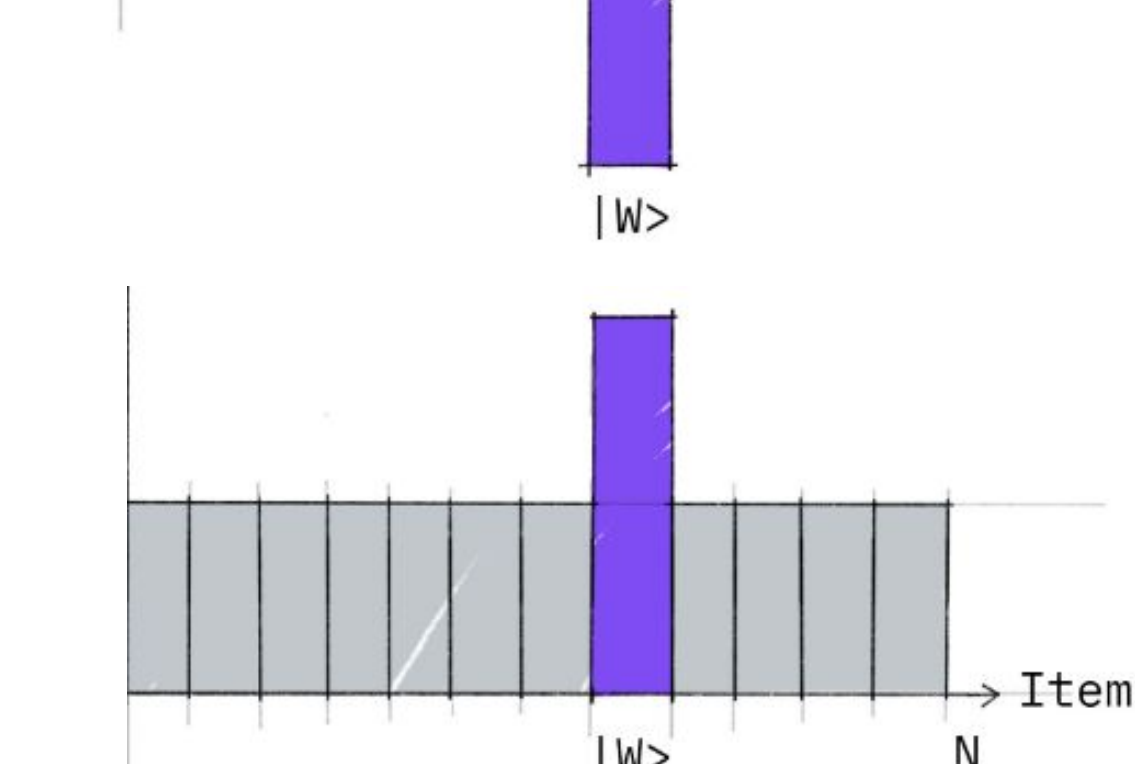


Figure 7

3. An Oracle reflection is applied to state S once again.

The process is complete. As shown in Figure 6, the target element is well distinguished from the remaining elements as the target has much higher probability when searching.

Results and Future Work

Running Sudoku Solver outputs 2 bit strings that match the qubits v_0-v_1 in Figure 1 that are the solutions. What we take away from this is a deeper meaning of Grover's Algorithm. Grover's algorithm can be used optimally in programming scenarios due to its efficiency, and can even be used to crack encryption algorithms. We plan on developing an algorithm that will fulfill these tasks to its fullest capability.